Smart Traffic Accident Analysis

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Contents

1 Introduction

This project analyzes 20,000 road accident records from Indian cities.

The analysis examines accident patterns based on time, location, weather, traffic density, severity, causes, casualties and risk scores. Machine learning models were also used to predict accident severity.

2 Dataset Overview

The dataset contains 20,000 accident records and 24 original variables. Additional analytical variables were created during the analysis.

2.1 Dataset Summary

- Total accidents: 20,000
- Cities: 8
- States: 7
- Years covered: 2022 to 2025
- Accident severity classes: Minor, Major and Fatal

3 Accident Severity

The accident severity distribution was:

- Minor accidents: 11,025
- Major accidents: 5,988
- Fatal accidents: 2,987

4 City Analysis

Chandigarh recorded the highest number of accidents with 2,577 accidents.

5 Accident Patterns by Hour

Accidents were analyzed across all 24 hours of the day.

6 Accident Patterns by Year

The dataset contains accident records from 2022, 2023, 2024 and 2025.

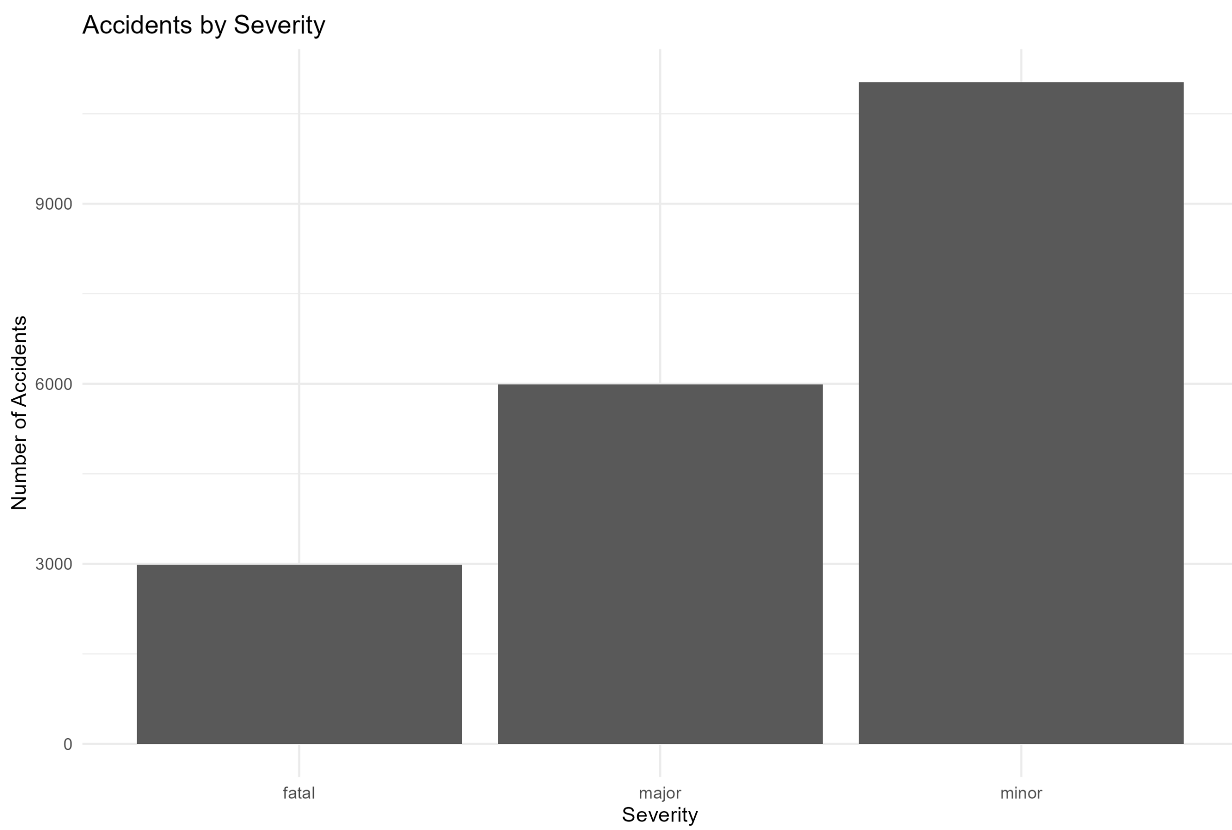

Figure 1: Accident Severity

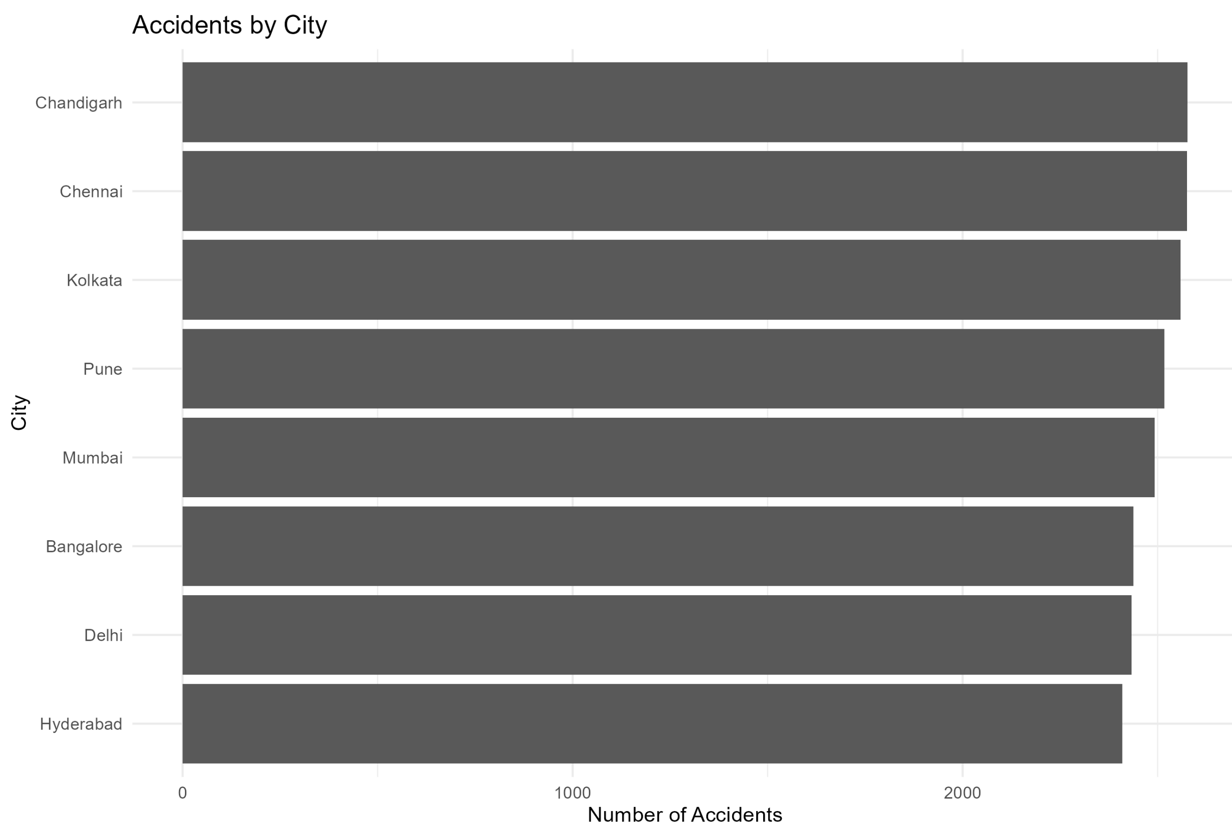

Figure 2: Accidents by City

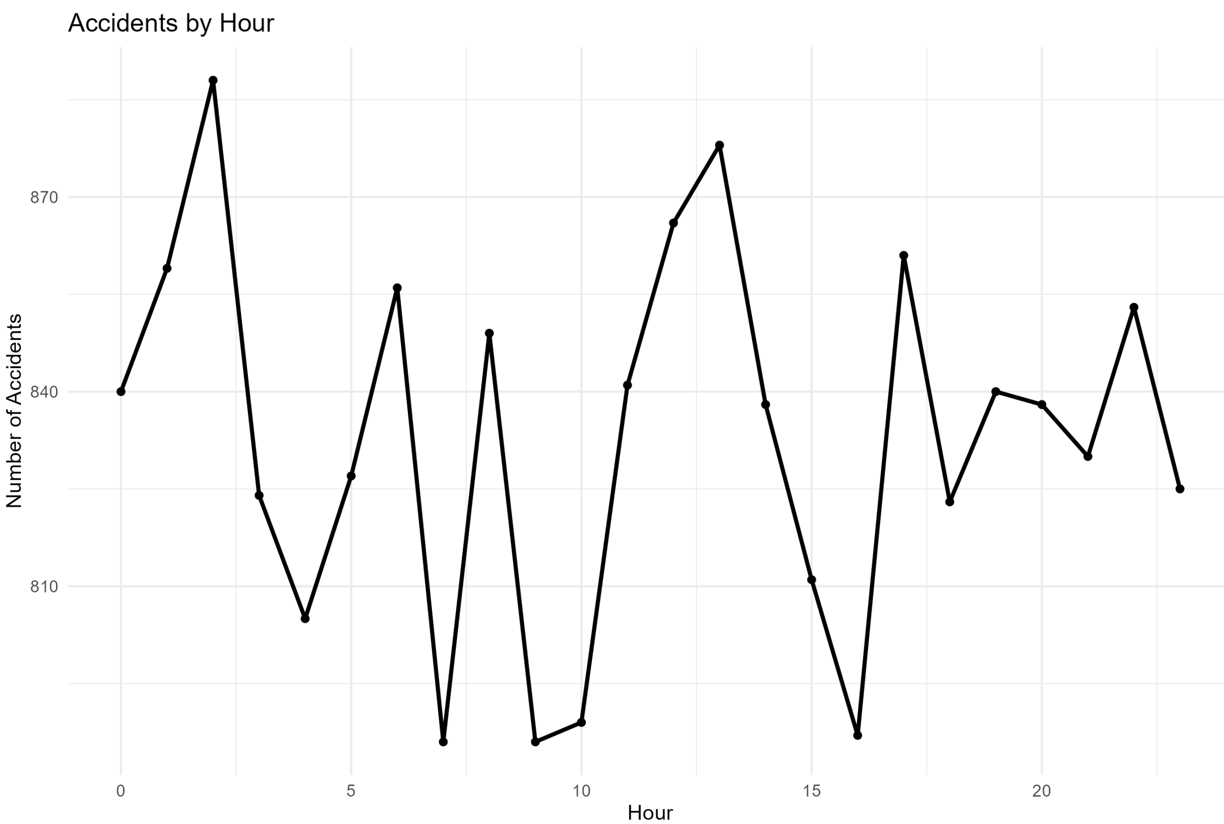

Figure 3: Accidents by Hour

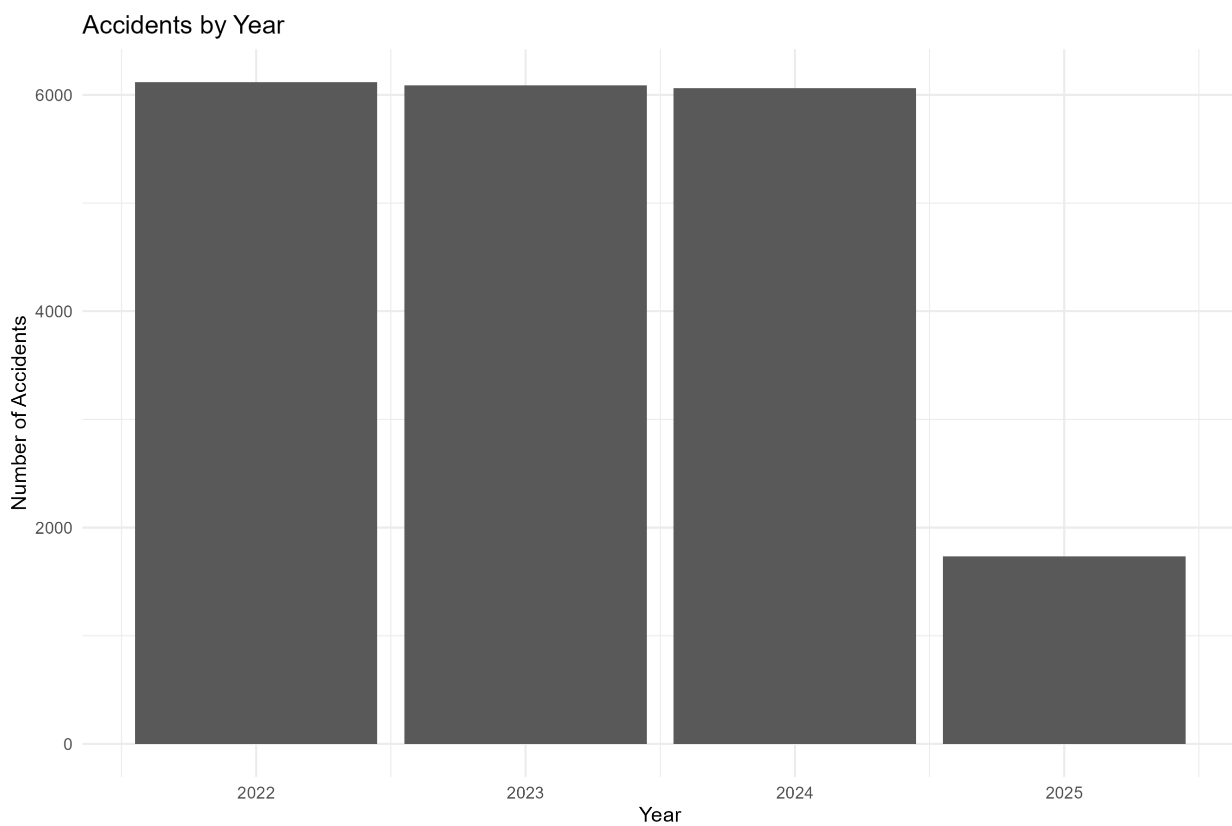

Figure 4: Accidents by Year

7 Weather Analysis

Fog had the highest average risk score:

- Fog: 0.589
- Rain: 0.488
- Clear: 0.237

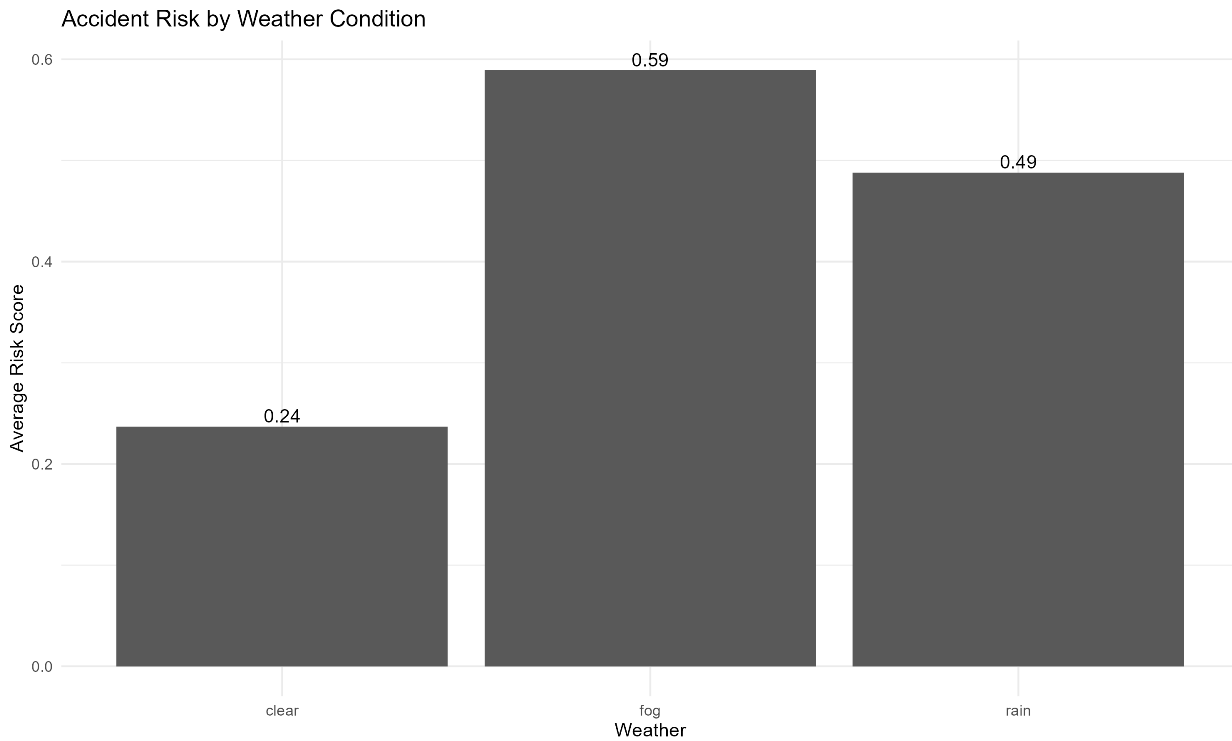

Figure 5: Weather Risk

8 Traffic Density Analysis

High traffic density had the highest average risk score:

- High: 0.595
- Medium: 0.368
- Low: 0.339

9 Accident Causes

The main accident causes included overspeeding, poor road conditions, weather, drunk driving and distraction.

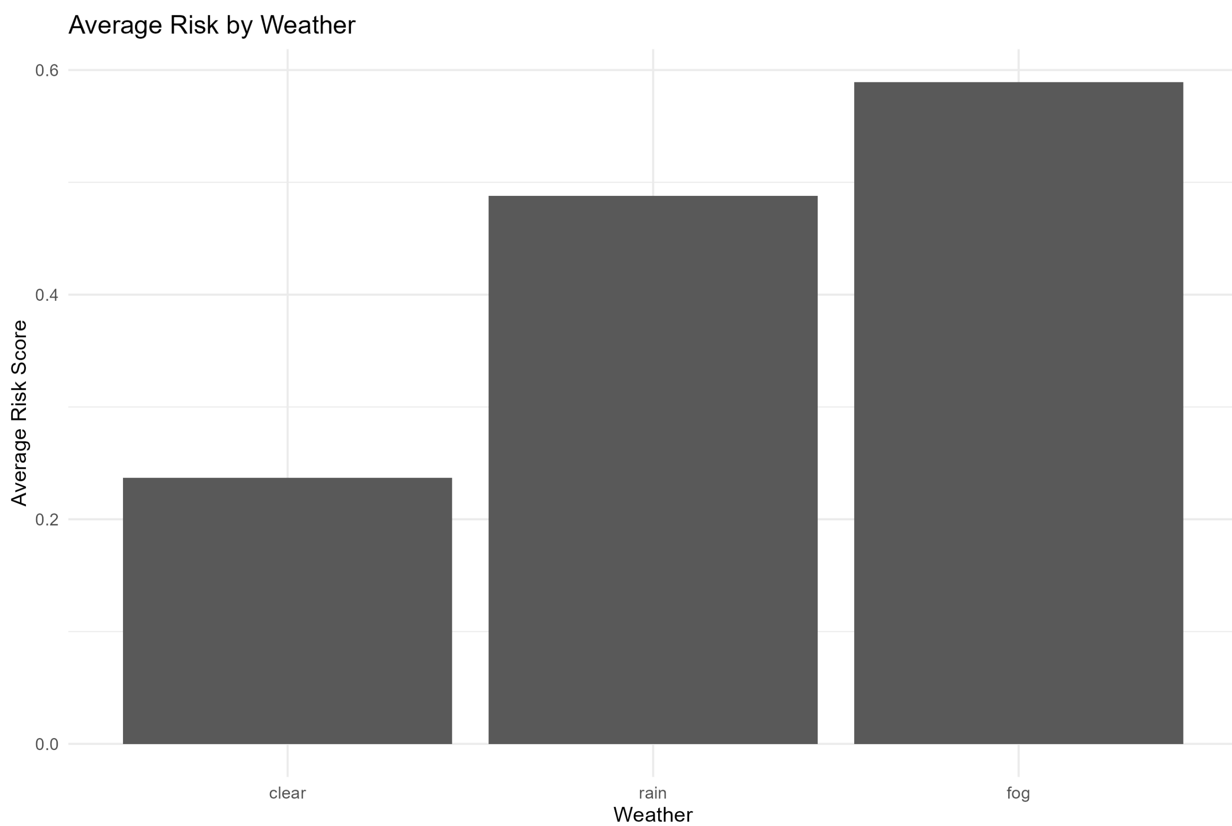

Figure 6: Traffic Risk

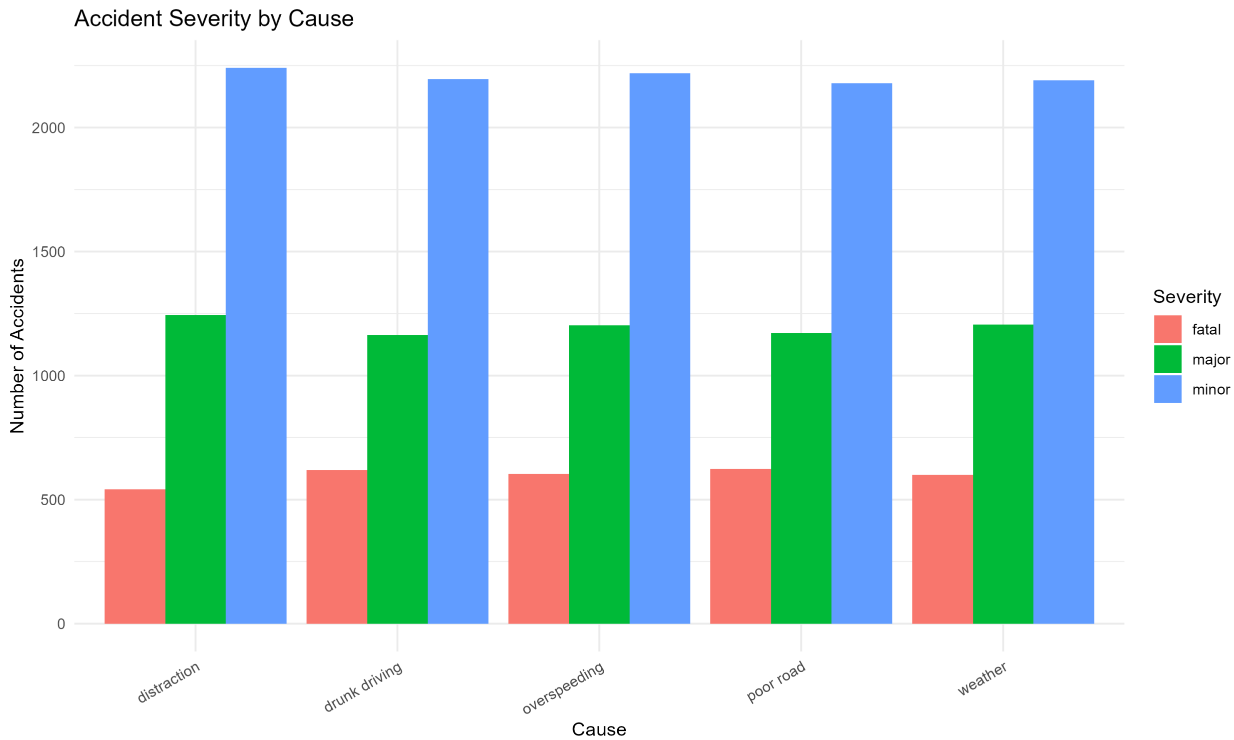

Figure 7: Cause and Severity

10 Traffic Hotspots

Spatial analysis identified areas with high concentrations of accidents.

11 Risk Analysis

The highest-risk combination was:

Fog + High Traffic + Fatal Accident

Average risk score: **0.919**

12 Severity by Hour

Accident severity was also examined across different hours of the day.

13 Machine Learning

Two machine learning models were evaluated:

- Random Forest
- Logistic Regression

The Random Forest model achieved an accuracy of **68.45%**.

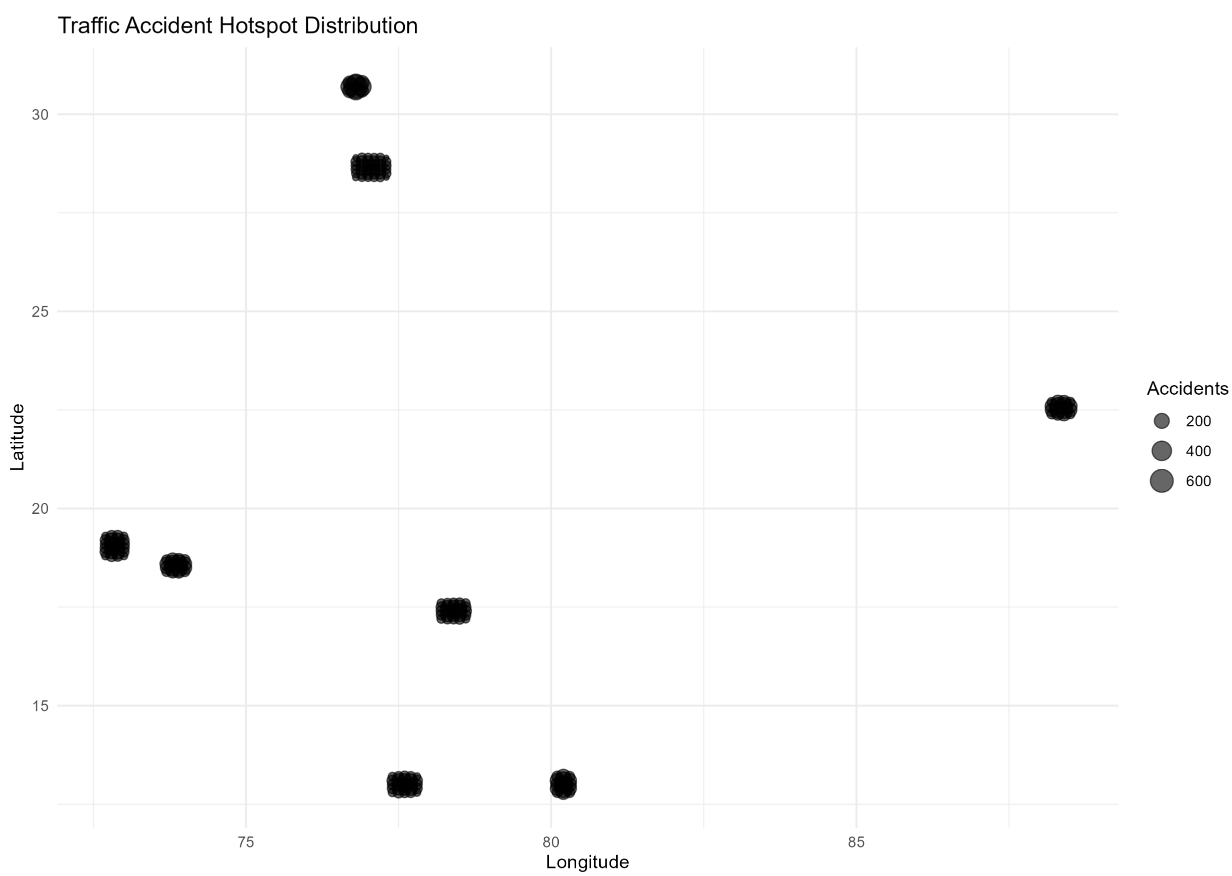

Figure 8: Traffic Hotspots

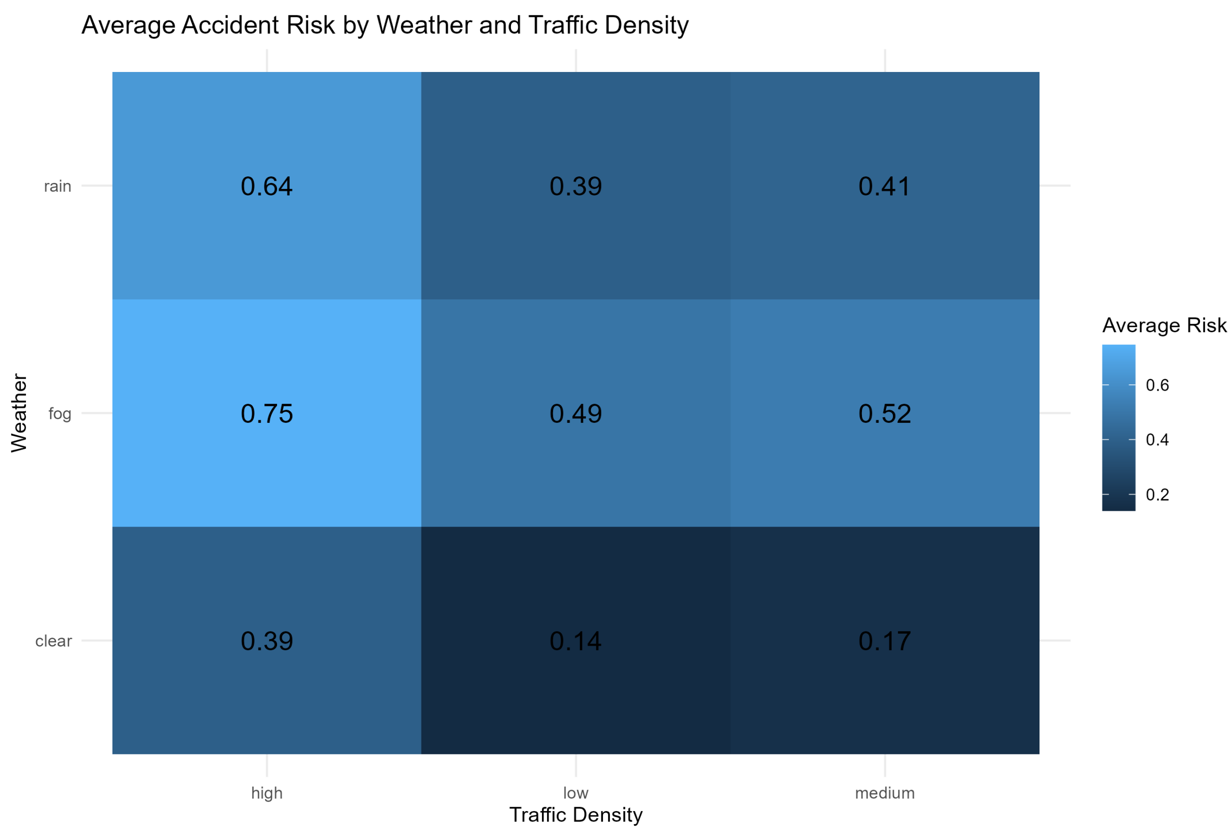

Figure 9: Weather Traffic Risk

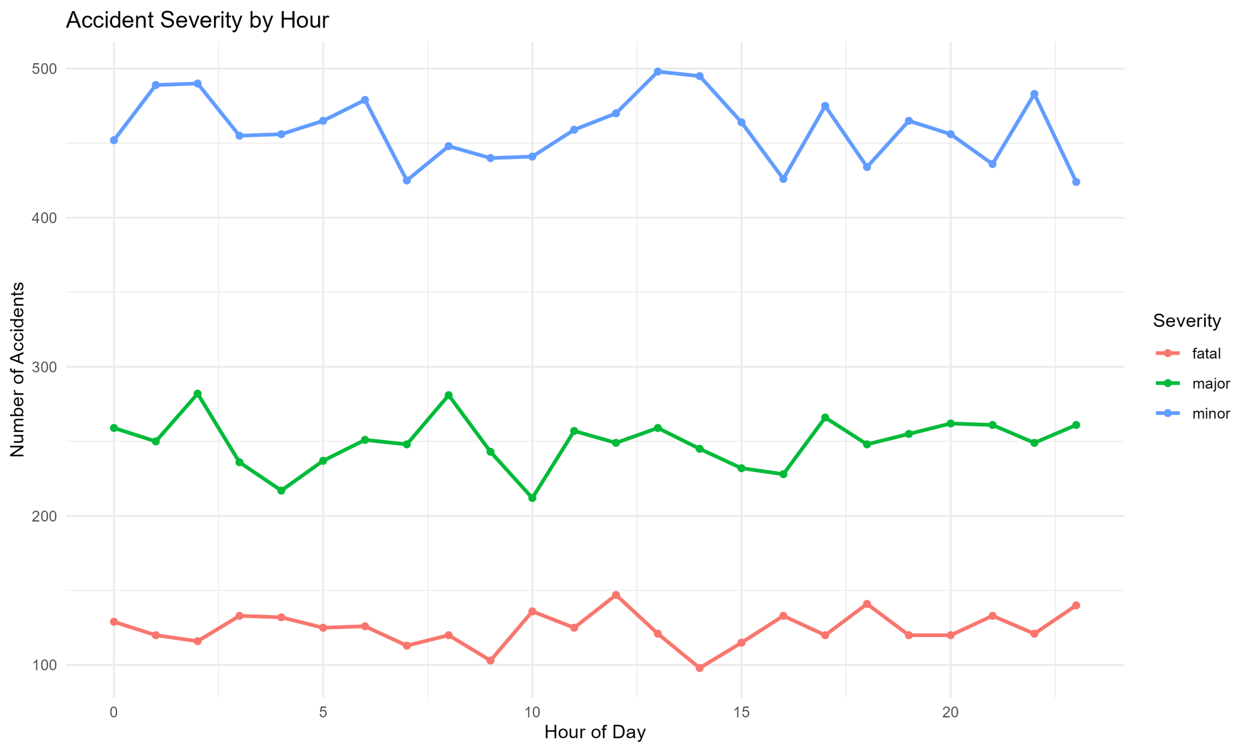

Figure 10: Severity by Hour

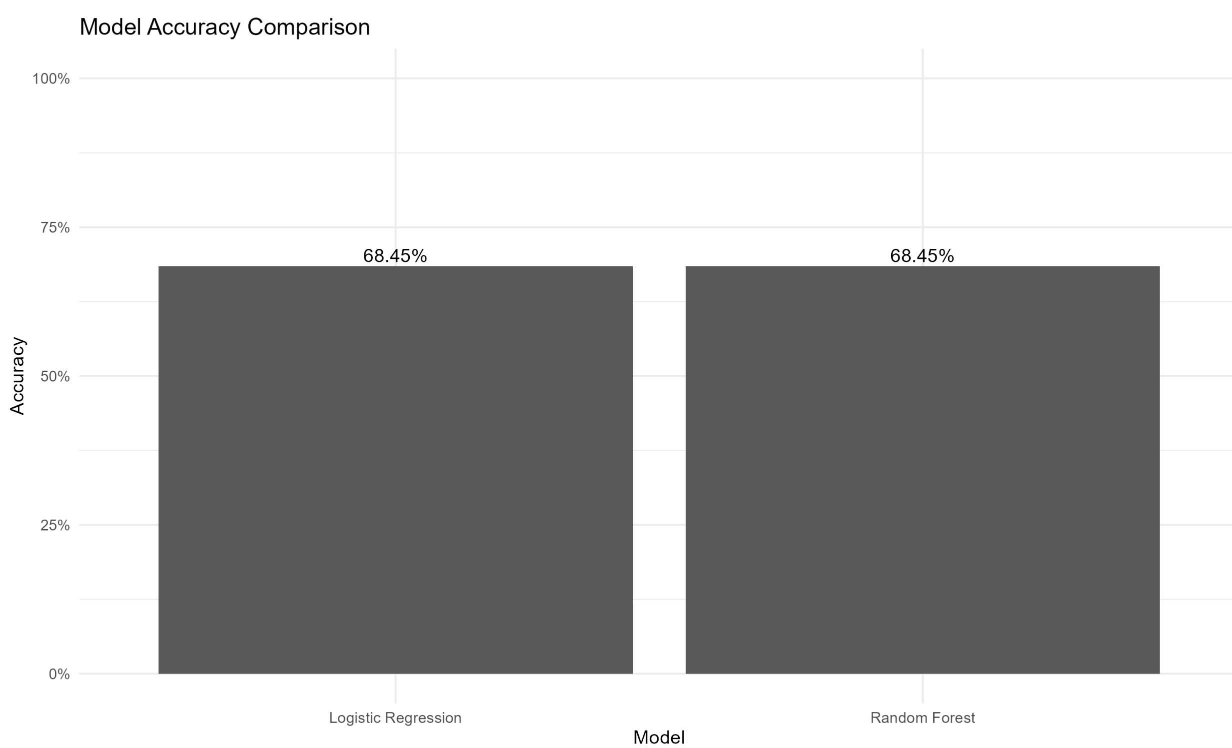

Figure 11: Model Comparison

14 Random Forest Performance

The Random Forest model used 300 trees.

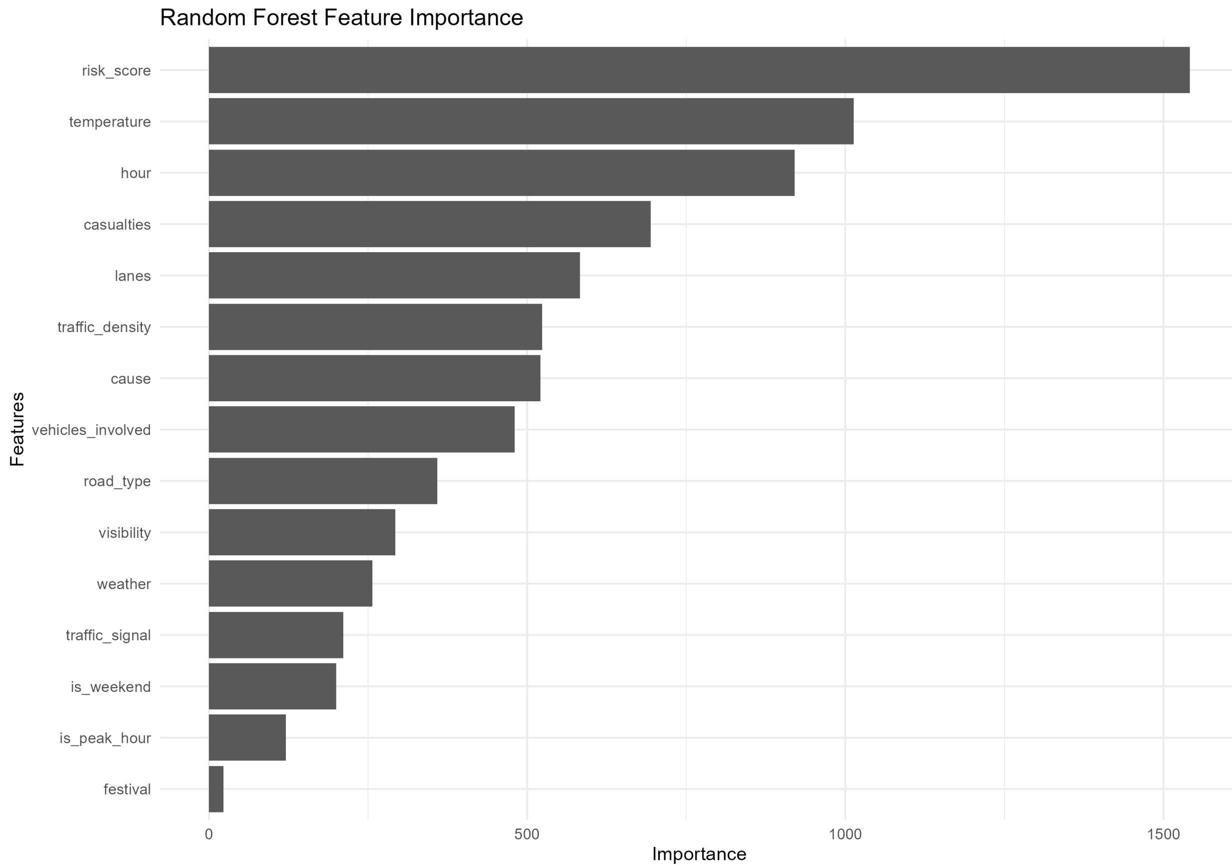

Figure 12: Feature Importance

The Random Forest confusion matrix is shown below.

15 Class Performance

The classification results were:

Severity	Precision	Recall	F1 Score
Fatal	99.83%	98.82%	99.32%
Major	39.64%	3.57%	6.56%
Minor	63.84%	96.88%	76.96%

Fatal accidents were identified very effectively. Minor accidents had high recall, while major accidents were difficult to classify accurately.

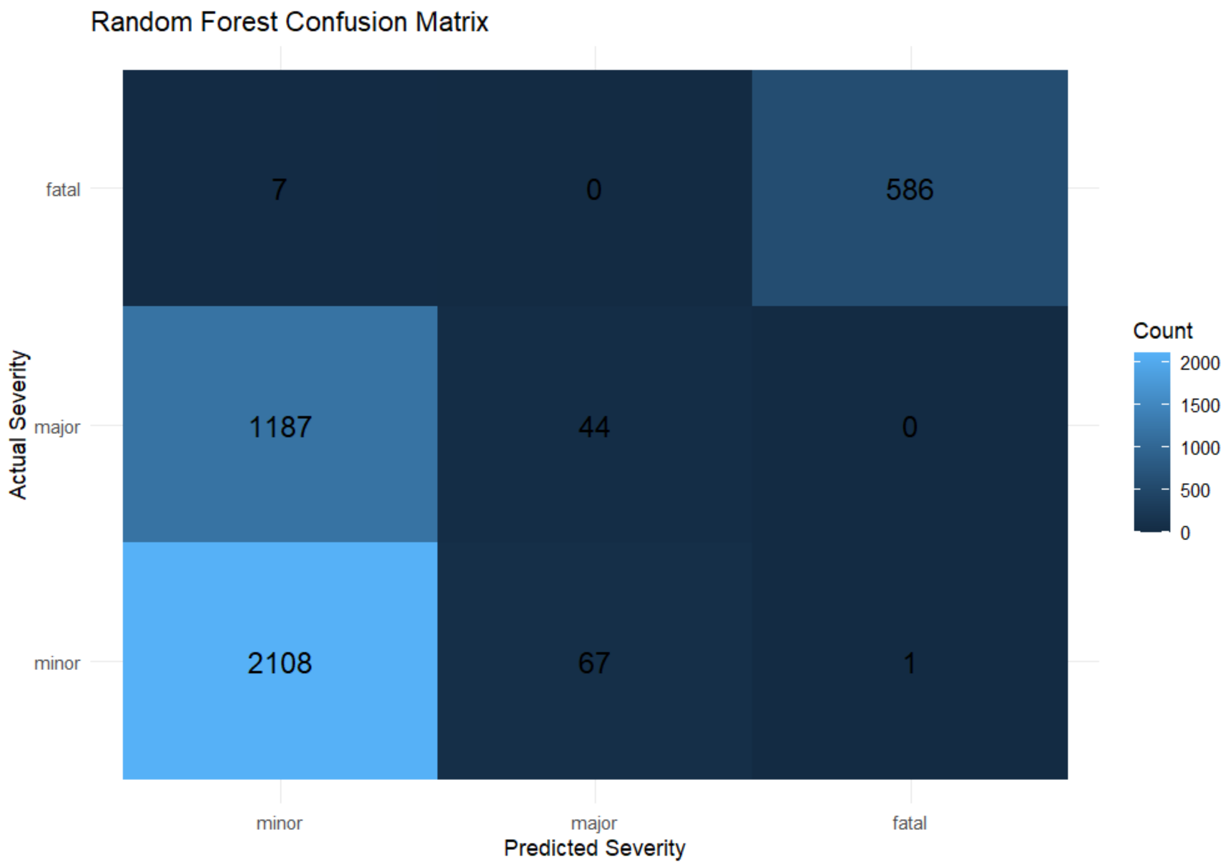

Figure 13: Random Forest Confusion Matrix

16 Conclusion

The analysis shows that weather and traffic density have strong relationships with accident risk.

Fog conditions and high traffic density were associated with higher risk. The highest-risk condition identified was fog combined with high traffic and fatal accident severity.

The machine learning models achieved an accuracy of 68.45%. Fatal accidents were classified very effectively, while major accidents remained difficult to distinguish accurately.

Overall, the analysis demonstrates how accident data, risk analysis and machine learning can be used to understand road safety patterns and identify high-risk conditions.